Chapter 2

Simple Random Sampling

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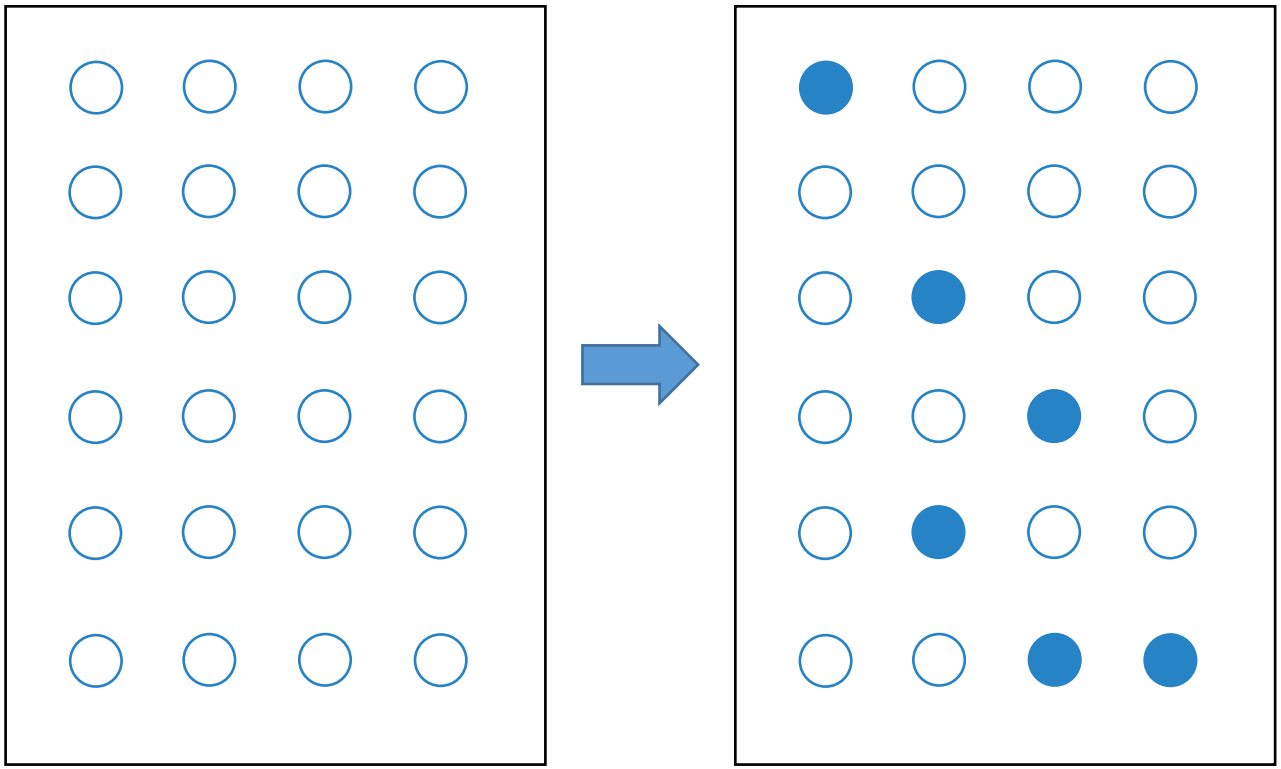
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Simple random sampling (SRS)

Motivation:

Choose n units from N units **without** replacement.



- Every one of the C_N^n distinct samples has an **equal** chance of being selected.
- The probability that any one of the C_N^n samples is selected is $1/C_N^n$.
- Draw-by-draw procedure
  - Consider $U = \{1, 2, \ldots, n\}$ and sample size n
  - In the first draw, select one element with equal probability.
  - In the second draw, select one element with equal probability from $U \setminus \{a_1\}$, where a_1 is the element selected from the first draw. Let a_2 be the element selected from the second draw.
  - And so on until n elements are drawn.

- Verify: all C_N^n distinct samples have an equal chance of being selected by the Draw-by-draw method.

  The probability that all n specified units are selected in n draws is

$$\frac{n}{N}\cdot\frac{(n-1)}{(N-1)}\cdot\frac{(n-2)}{(N-2)}\cdots\frac{1}{(N-n+1)}=\frac{n!(N-n)!}{N!}=\frac{1}{C_N^n}$$

- Use random number tables, computers

- In SRS, all units in the population have the same fixed selection probability of n/N.

Definitions and notations

- The properties of interest: characteristics or items
- Population size: N Sample size: n
- The values for any specific item in the population:

$$y_1, \ldots, y_N \quad \text{or} \quad Y_1, \ldots, Y_N$$

- The corresponding values for the units in the sample:

$$y_1, y_2 \ldots, y_n$$

Total and mean

	Population	Sample
Total	$Y = \sum_{i=1}^{N} y_i = y_1 + y_2 + \cdots + y_N$	$y = \sum_{i=1}^{n} y_i = y_1 + y_2 + \cdots + y_n$
Mean	$\bar{Y} = \frac{y_1 + y_2 + \cdots + y_N}{N} = \frac{\sum_{i=1}^{N} y_i}{N}$	$\bar{y} = \frac{y_1 + y_2 + \cdots + y_n}{n} = \frac{\sum_{i=1}^{n} y_i}{n}$

Four most interested characteristics of the population

1. Mean: $\bar{Y}$ 2. Total: Y

3. Ratio of two totals or means: $R = Y / X = \bar{Y} / \bar{X}$

4. Proportion of units that fall into some defined class: $P = \frac{A}{N}$

Simplest estimators

Population value	Estimator
Population mean $\bar{Y}$	$\hat{\bar{Y}} = \bar{y} = \text{sample mean}$
Population total Y	$\hat{Y} = N\bar{y} = N\sum^{n} y_i/n$
Population ratio R	$\hat{R} = \bar{y}/\bar{x} = \sum^{n} y_i / \sum^{n} x_i$
Population proportion P	$p = a/n$

N / n: expansion or raising or inflation factor

$f = n / N$: sampling fraction

Properties of the estimates

For a method of estimation,

- consistent: the estimate becomes exactly equal to the population value when $n = N$.
- unbiased: the average value of the estimate, taken over all possible samples of size n, is exactly equal to the true population value.

Theorem 2.1 The sample mean $\bar{y}$ is an unbiased estimate of $\bar{Y}$. That is $E(\bar{y}) = \bar{Y}$.

Proof

Corollary $\hat{Y} = N\bar{y}$ is an unbiased estimate of the population total Y.

A fact is

$E(y_1 + y_2 + \cdots + y_n)$ must be n/N multiple of $(y_1 + y_2 + \cdots + y_N)$.

- The LHS has n terms, and the RHS has N terms.
- Every unit appears in the same number of samples.

Variances of the estimates

Different definitions of the variance in a finite population

- Usual definition

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(y_i - \bar{Y})^2$$

- The definition in sampling theory

$$S^2 = \frac{1}{N-1}\sum_{i=1}^{N}(y_i - \bar{Y})^2 = \frac{N}{N-1}\sigma^2$$

- Advantage of using S^2 : most results taking a slightly simpler form

Theorem 2.2 The variance of the mean $\bar{y}$ from a simple random sample is

$$V(\bar{y}) = E(\bar{y} - \bar{Y})^2 = \frac{S^2}{n}(1 - f) = \frac{S^2}{n}\frac{(N-n)}{N}$$

Proof

Corollary 1. The standard error of $\bar{y}$ is

$$\sigma_{\bar{y}} = \frac{S}{\sqrt{n}}\sqrt{1-f} = \frac{S}{\sqrt{n}}\sqrt{\frac{N-n}{N}}$$

Corollary 2. The variance of $\hat{Y} = N\bar{y}$, as an estimate of the population total Y , is

$$V(\hat{Y}) = N^2 V(\bar{y}) = \frac{N^2 S^2}{n}(1-f) = \frac{N^2 S^2}{n}\frac{N-n}{N}$$

Corollary 3. The standard error of $\hat{Y}$ is

$$\sigma_{\hat{Y}} = \frac{NS}{\sqrt{n}}\sqrt{1-f} = \frac{NS}{\sqrt{n}}\sqrt{\frac{N-n}{N}}$$

Random sampling with replacement

Choose n units from N units **with** replacement.

- According to the knowledge in mathematical statistics,

$$E(\bar{y}) = \bar{Y}$$

$$V(\bar{y}) = \frac{\sigma^2}{n} = \frac{1}{nN}\sum_{i=1}^{N}(y_i - \bar{Y})^2 = \frac{N-1}{N}\frac{S^2}{n}$$

- $V_{\text{srs}}(\bar{y}) = \dfrac{N-n}{N}\dfrac{S^2}{n}$ in sampling without replacement is only $\boxed{(N-n)/(N-1)}$ times the variance in sampling with replacement.
- They are equal to each other when $n = 1$.

The finite population correction

Finite population correction (fpc)

- $(N-n)/N = 1-f$ for the variance
- $\sqrt{(N-n)/N} = \sqrt{1-f}$ for the standard error

- In practice, the fpc can be ignored whenever $f \leq 5\%$ and for many purpose even if $f \leq 10\%$.
- Ignore the fpc $\rightarrow$ overestimate $\sigma_{\bar{y}}$

Estimation of the standard error from a sample

Three primary uses of the formulas for $\sigma_{\bar{y}}$, $\sigma_{\hat{Y}}$

- to compare the precision obtained by SRS with that given by other methods of sampling
- to estimate the sample size need in a survey that is being planned
- to estimate the precision actually attained in a survey that has been completed

Problem: S^2 is unknown ⇨ Solution: to estimate

Theorem 2.4 For a simple random sample, sample variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2$$

is an unbiased estimate of $S^2 = \sum_{i=1}^{N}(y_i - \bar{Y})^2/(N-1)$.

Proof:

Corollary Unbiased estimates of the variances of $\bar{y}$ and $\hat{Y} = N\bar{y}$ are

$$v(\bar{y}) = s^2_{\bar{y}} = \frac{s^2}{n}\frac{N-n}{N} = \frac{s^2}{n}(1-f)$$

$$v(\hat{Y}) = s^2_{\hat{Y}} = \frac{N^2 s^2}{n}\frac{N-n}{N} = \frac{N^2 s^2}{n}(1-f)$$

For the standard errors, the estimates are

$$s_{\bar{y}} = \frac{s}{\sqrt{n}}\sqrt{1-f} \qquad s_{\hat{Y}} = \frac{Ns}{\sqrt{n}}\sqrt{1-f}$$

* The above two estimates are slightly biased: for most applications the bias in unimportant.

Confidence limits

Assume the estimates $\bar{y}$, $\hat{Y}$ are normally distributed.

- Mean

$$\hat{\bar{Y}}_L = \bar{y} - \frac{ts}{\sqrt{n}}\sqrt{1-f} \qquad \hat{\bar{Y}}_U = \bar{y} + \frac{ts}{\sqrt{n}}\sqrt{1-f}$$

- Total

$$\hat{Y}_L = N\bar{y} - \frac{tNs}{\sqrt{n}}\sqrt{1-f} \qquad \hat{Y}_U = N\bar{y} + \frac{tNs}{\sqrt{n}}\sqrt{1-f}$$

Confidence probability (%)	50	80	90	95	99
t	0.67	1.28	1.64	1.96	2.58



Example

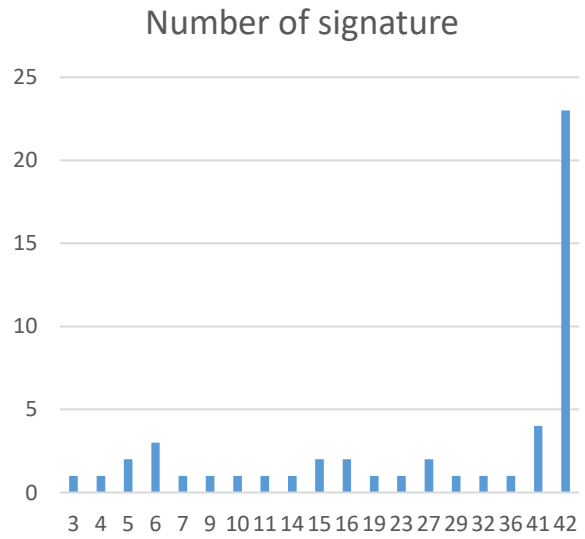
Signatures to a petition were collected on 676 sheets. Each sheet had enough space for 42 signatures, but on many sheets a smaller number of signatures had been collected. The numbers of signatures per sheet were counted on a random sample of 50 sheets (about a 7% sample), with the results shown in Table 2.2.

Estimate the total number of signatures to the petition and the 80% confidence limits.

Table 2.2 Results for a sample of 50 petition sheets

y_i'	42	41	36	32	29	27	23	19	16	15
f_i	23	4	1	1	1	2	1	1	2	2
y_i'	14	11	10	9	7	6	5	4	3	Total
f_i	1	1	1	1	1	3	2	1	1	50

$y_i' =$ number of signatures; $f_i =$ frequency



$$n = \sum f_i = 50 \quad y = \sum_{i=1}^{n} y_i = \sum f_i y_i' = 1471$$

$$\sum_{i=1}^{n} y_i^2 = \sum f_i y_i'^2 = 54497$$

Estimation of a ratio

Interested: the ratio of two variables both of which vary from unit to unit, from a simple random sample



Example In a household survey, we are interested in the following three quantities.

- The average number of suits of clothes per adult male;
- The average expenditure on cosmetics per adult female;
- The average number of hours per week spent watching television per child aged 10 to 15.

Need to observe: (1) the number of adult male, and (2) the number of suits of clothes in a selected family

For the ith unit, we collect a pair of observations (x_i, y_i).

- The population parameter to be estimated is the ratio

$$R = \frac{\sum_{i=1}^{N} y_i}{\sum_{i=1}^{N} x_i} = \frac{Y}{X} = \frac{\overline{Y}}{\overline{X}}$$

- The corresponding sample estimate is

$$\hat{R} = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i} = \frac{\overline{y}}{\overline{x}}$$

- The sampling distribution of $\hat{R}$ is more complicated than that of $\bar{y}$.
- In small samples
  - the distribution of $\hat{R}$ is skew;
  - $\hat{R}$ is usually a slightly biased estimate of R .
- In large sample,
  - the distribution of $\hat{R}$ tends to normality;
  - the bias becomes negligible .

Theorem 2.5 If variates (y_i, x_i) are measured on each unit of a simple random sample of size n, assumed large, the MSE and variance of $\hat{R} = \bar{y} / \bar{x}$ are each approximately

$$V(\hat{R}) \doteq \text{MSE}(\hat{R}) = \frac{1-f}{n\bar{X}^2} \frac{\sum_{i=1}^{N} (y_i - Rx_i)^2}{N-1}$$

where $R = \bar{Y} / \bar{X}$ is the ratio of the population means and $f = n / N$.

Proof

- The variance of sample ratio $\hat{R}$

$$V(\hat{R}) \doteq \mathrm{MSE}(\hat{R}) = \frac{1-f}{n\bar{X}^2} \frac{\sum_{i=1}^{N}(y_i - Rx_i)^2}{N-1}$$

- The variance of the sample mean $\bar{y}$

$$V(\bar{y}) = \frac{1-f}{n} \frac{\sum_{i=1}^{N}(y_i - \bar{Y})^2}{N-1}$$

- Replace the variate y_i by $(y_i - Rx_i)/\bar{X}$

- Estimate:

$$\frac{\sum_{i=1}^{n}(y_i - \hat{R}x_i)^2}{n-1} \quad \Rightarrow \quad \frac{\sum_{i=1}^{N}(y_i - Rx_i)^2}{N-1}$$

The estimate has a bias of order $1/n$.

$$V(\hat{R}) \doteq \mathrm{MSE}(\hat{R}) = \frac{1-f}{n\bar{X}^2} \frac{\sum_{i=1}^{N}(y_i - Rx_i)^2}{N-1} + \frac{\sum_{i=1}^{n}(y_i - \hat{R}x_i)^2}{n-1}$$

- The estimated standard error of $\hat{R}$

$$S(\hat{R}) \doteq \frac{\sqrt{1-f}}{\sqrt{n}\bar{X}} \sqrt{\frac{\sum_{i=1}^{n}(y_i - \hat{R}x_i)^2}{n-1}}$$

- Another expression:

$$s(\hat{R}) \doteq \frac{\sqrt{1-f}}{\sqrt{n}\bar{X}} \sqrt{\frac{\sum_{i=1}^{n} y_i^2 - 2\hat{R}\sum_{i=1}^{n} y_i x_i + \hat{R}^2 \sum_{i=1}^{n} x_i^2}{n-1}}$$

- If $\bar{X}$ is not known, the sample mean $\bar{x}$ can be used.



Example

Table 2.3 shows the number of persons (x_1), the weekly family income (x_2), and the weekly expenditure on food (y) in a simple random sample of 33 low-income families. Since the sample is small, the data are intended only to illustrate the calculations.

Estimate from the sample

(a) the mean weekly expenditure on food per family,

(b) the mean weekly expenditure on food per person, and

(c) the percentage of the income that is spent on food.

Compute the standard errors of these estimates.

TABLE 2.3

SIZE, WEEKLY INCOME, AND FOOD COST OF 33 FAMILIES

Family Number	Size x_1	Income x_2	Food Cost y	Family Number	Size x_1	Income x_2	Food Cost y
1	2	62	14.3	18	4	83	36.0
2	3	62	20.8	19	2	85	20.6
3	3	87	22.7	20	4	73	27.7
4	5	65	30.5	21	2	66	25.9
5	4	58	41.2	22	5	58	23.3
6	7	92	28.2	23	3	77	39.8
7	2	88	24.2	24	4	69	16.8
8	4	79	30.0	25	7	65	37.8
9	2	83	24.2	26	3	77	34.8
10	5	62	44.4	27	3	69	28.7
11	3	63	13.4	28	6	95	63.0
12	6	62	19.8	29	2	77	19.5
13	4	60	29.4	30	2	69	21.6
14	4	75	27.1	31	6	69	18.2
15	2	90	22.2	32	4	67	20.1
16	5	75	37.7	33	2	63	20.7
17	3	69	22.6				
				Total	123	2394	907.2

- Weekly expenditure on food per family

$$\bar{y} = \frac{907.2}{33} = 27.49$$

$$v(\bar{y}) = \frac{1-f}{n} s_y^2 = \frac{1}{n} \frac{\sum_{i=1}^{n} (y_i - \bar{y})^2}{n-1} = \frac{3284.2473}{33 \times 32} = 3.11$$

$$se(\bar{y}) = 1.76$$

- Weekly expenditure on food per person

$$\hat{R}_1 = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_{1i}} = \frac{907.2}{123} = 7.38$$

$$se(\hat{R}_1) = \frac{\sqrt{1-f}}{\sqrt{n}\bar{x}_1} \sqrt{\frac{\sum_{i=1}^{n} y_i^2 - 2\hat{R}_1 \sum_{i=1}^{n} y_i x_{1i} + \hat{R}_1^2 \sum_{i=1}^{n} x_{1i}^2}{n-1}}$$

$$= \frac{1}{\sqrt{33} \times (123/33)} \sqrt{\frac{28224 - 2 \times 7.38 \times 3595.5 + 7.38^2 \times 533}{32}} = 0.534$$

- Percentage of income spent on food

$$\hat{R}_2 = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_{2i}} \times 100\% = \frac{907.2}{2394} \times 100\% = 37.9\%$$

$$se(\hat{R}_2) = \frac{\sqrt{1-f}}{\sqrt{n}\bar{x}_2} \sqrt{\frac{\sum_{i=1}^{n} y_i^2 - 2\hat{R}_2 \sum_{i=1}^{n} y_i x_{2i} + \hat{R}_2^2 \sum_{i=1}^{n} x_{2i}^2}{n-1}}$$

$$= \frac{1}{\sqrt{33} \times (2394/33)} \sqrt{\frac{28224 - 2 \times 0.379 \times 177254 + 0.378^2 \times 66678}{32}}$$

$$= 0.0238$$

Theorem 2.3 If y_i, x_i are a pair of variates defined on every unit in the population and $\bar{y}, \bar{x}$ are the means from a simple random sample of size n, then their covariance

$$\begin{aligned}\mathrm{Cov}(\bar{y}, \bar{x}) &= E(\bar{y} - \bar{Y})(\bar{x} - \bar{X}) \\ &= \frac{N-n}{nN}\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \bar{Y})(x_i - \bar{X}) = \frac{1-f}{n}S_{yx}\end{aligned}$$

where $S_{yx} = \frac{1}{N-1}\sum_{i=1}^{N}(y_i - \bar{Y})(x_i - \bar{X})$ is the population covariance.

Proof

Estimation of means over subpopulation

Domains of study: a number of classes into which the population is subdivided, or subpopulations

Suppose

- Each unit falls into one of the domains
- The jth domain contains N_j units
- n_j : the number of units in a simple random sample of size n that happen to fall in this domain
- y_{jk} $(k = 1, 2, \cdots, n_j)$: the measurements of the n_j units
- The population mean $\bar{Y}_j$

- The estimator is the sample mean for the *j*th domain

$$\bar{y}_j = \frac{\sum_{k=1}^{n_j} y_{jk}}{n_j}$$

- Is it a ratio estimate?
- Consider both n and n_j are fixed. Assume $n_j > 0$.
- The n_j units can be treated as a simple random sample selected from a population of size N_j .

$$\frac{C_{N-N_j}^{n-n_j}}{C_{N-N_j}^{n-n_j} C_{N_j}^{n_j}} = \frac{1}{C_{N_j}^{n_j}}$$

- $\bar{y}_j$ is an unbiased estimate of $\bar{Y}_j$.
- The standard error of $\bar{y}_j$ is

$$S(\bar{y}_j) = E(\bar{y}_j - \bar{Y}_j)^2 = \sqrt{\frac{1-f_j}{n_j}} S_j = \sqrt{\frac{N_j - n_j}{n_j N_j}} S_j$$

where

$$S_j^2 = \frac{\sum_{k=1}^{N_j} (y_{ik} - \bar{Y}_j)^2}{N_j - 1}$$

- An estimate of the standard error of $\bar{y}_j$ is

$$s(\bar{y}_j) = \sqrt{\frac{1-f_j}{n_j}} s_j$$

where $s_j^2 = \frac{\sum_{k=1}^{n_j} (y_{ik} - \bar{y}_j)^2}{n_j - 1}$ and $\hat{f}_j = f$.

Estimation of totals over subpopulation

- If N_j is known, the sample estimate is $N_j \bar{y}_j$.
- Otherwise…

- Recall the raising factor N / n
- The estimate for the total of the jth domain is

$$\hat{Y}_j = \frac{N}{n} \sum_{k=1}^{n_j} y_{jk}$$

- An unbiased estimator

Define $$y_i' = \begin{cases} y_i, & \text{if the unit } i \text{ is in the } j\text{th domain} \\ 0, & \text{otherwise} \end{cases}$$

The population total of the y_i' is

$$\sum_{i=1}^{N} y_i' = \sum_{j\text{th domain}} y_i = Y_j$$

Denote by $\bar{y}'$ the ordinary sample mean of the y_i'

$$N\bar{y}' = \frac{N}{n}\sum_{i=1}^{n} y_i' = \frac{N}{n}\sum_{i=1}^{n_j} y_{jk} = \hat{Y}_j$$

$$\sigma_{\hat{Y}_j} = \frac{NS'}{\sqrt{n}}\sqrt{1-f} = \frac{NS'}{\sqrt{n}}\sqrt{1-\frac{n}{N}}$$

That is $$S'^2 = \frac{\sum_{i=1}^{N}(y_i' - \bar{Y}')^2}{N-1} = \frac{1}{N-1}\left(\sum_{j\text{th domain}} y_i^2 - \frac{Y_j^2}{N}\right)$$

A sample estimate of the standard error of $\hat{Y}_j$ is

$$s_{\hat{Y}_j} = \frac{Ns'}{\sqrt{n}}\sqrt{1-f} = \frac{Ns'}{\sqrt{n}}\sqrt{1-\frac{n}{N}}$$

In computing s', any unit not in the jth domain is given 0.

* The methods also apply to surveys in which the frame used contains units that do not belong to the population.



Example

From a list of 2422 minor household expenditures a simple random sample of 180 items was drawn in order to estimate the total spent for operation of the household. Certain types of expenditure (on clothing and car upkeep) were not considered relevant. Of the 180 sample items, 152 were relevant. The sum and uncorrected sum of squares of the relevant amounts (in dollars) were as follows.

$$\sum y_i' = 343.5, \quad \sum y_i'^2 = 1491.38$$

Estimate the total expenditure for household operation and give the standard error of the estimate.

$$\sum y_i' = 343.5, \quad \sum y_i'^2 = 1491.38$$

Validity of the normal approximation

- We have learned “for any population that has a finite standard deviation, the distribution of the sample mean tends to normality as n increases”.
- This statement relates to infinite populations.
- For sampling without replacement from finite population, some necessary and sufficient conditions are needed, for which the distribution of sample mean tends to normality.
- Not easy to answer “For the population, how large must n be so that the normal approximation is accurate enough?”

- Non-normal distributions vary greatly both in the nature and in the degree of their departure from normality.

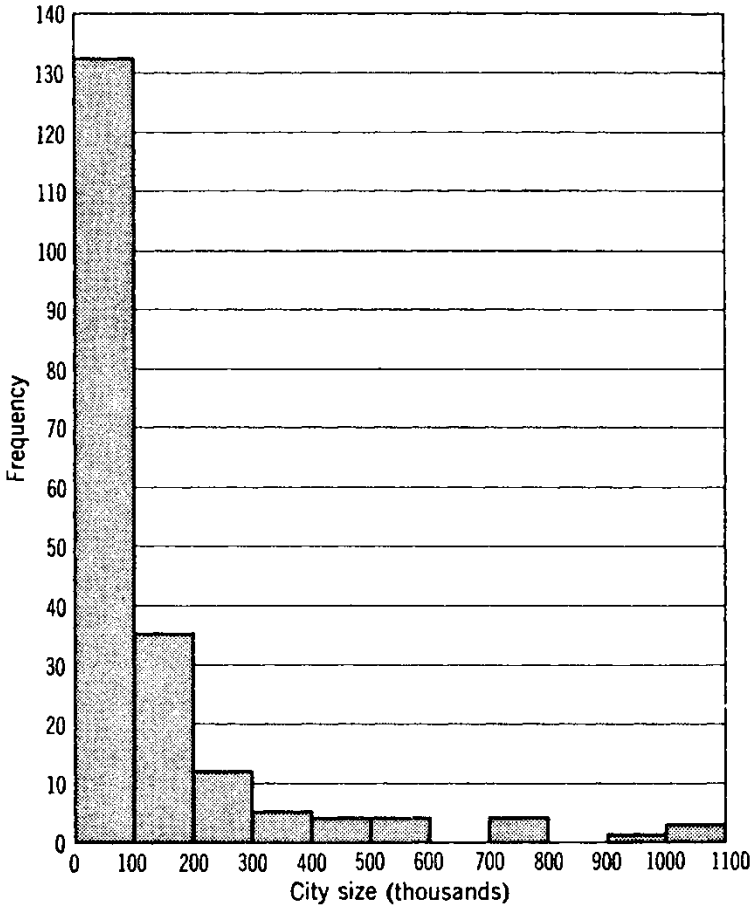


Fig. 2.1. Frequency distribution of sizes of 196 United States Cities in 1920.

- The frequency distribution of the total number of inhabitants in each of 200 simple random samples with $n = 49$

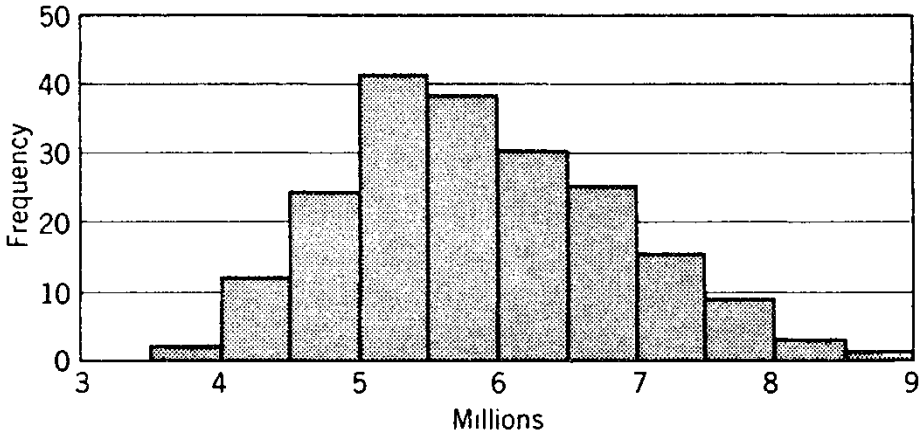


Fig. 2.2 Frequency distribution of totals of 200 simple random samples with n=49.

- Much more similar to a normal curve, but still with some positive skewness